

Invasive alien species

Ironhack DSMLFT September 2026

Project 1 EDA and SQL

What is an invasive alien species?

An **invasive alien species** is an organism, such as an animal or plant, that has **established** itself in a **new environment** where it is **not native**. Its introduction can be intentional or unintentional, through human activities or other means.



Why investigate invasive alien species?

Invasive species impact:

- Nature
 - ecosystem disruption
 - threatening native species
 - biodiversity loss
- Economy
 - lost of native species
 - change in activities
 - costly control
- Human health
 - spread of diseases
 - Destruction of living space

→ Understand to take action!

FEATURE | 30 September 2026

Invasive plants are pushing African communities from their land

Across rural Africa, invasive species are shrinking grazing and farming land, and damaging livelihoods.

By [Gilbert Nakweya](#)



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Invasive *Opuntia stricta* could cost sub-Saharan Africa an estimated US\$307 billion in lost livestock production over the next 50 years. Credit: Arne Witt, CABI

Global Impacts Dataset of Invasive Alien Species

Published 21. of Mai 2025 authored by S. Bacher et al.

<https://doi.org/10.6084/m9.figshare.27908838>

Each **row** represents a **record** of an invasive alien **species** at a specific **location**, including its **impact** on the environment and human activities, as well as the **year** in which the impact was recorded or published.

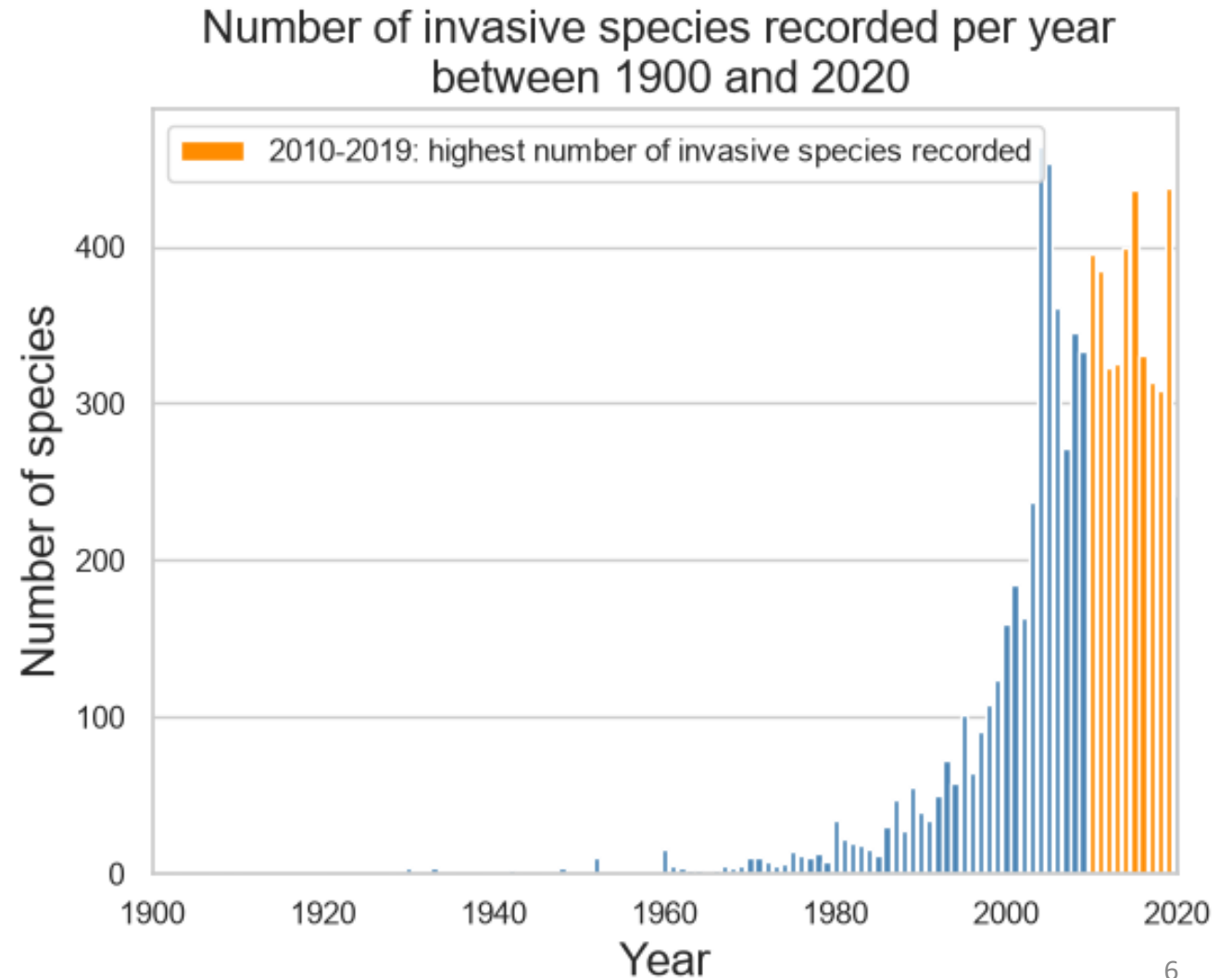
rowID	UniqueID	IAS.Species.Name	Verified.Name.GBIF.Taxon	GBIF.scientificName.with.author	Genus	Family	Order	Class	Phylum
0	1	AF_Fish_crayfish_00001	Gambusia affinis	Gambusia affinis	Gambusia affinis (Baird & Girard, 1853)	Gambusia	Poeciliidae	Cyprinodontiformes	NaN
1	2	AF_Fish_crayfish_00002	Onchorhynchus mykiss	Oncorhynchus mykiss	Oncorhynchus mykiss (Walbaum, 1792)	Oncorhynchus	Salmonidae	Salmoniformes	NaN

Research Questions

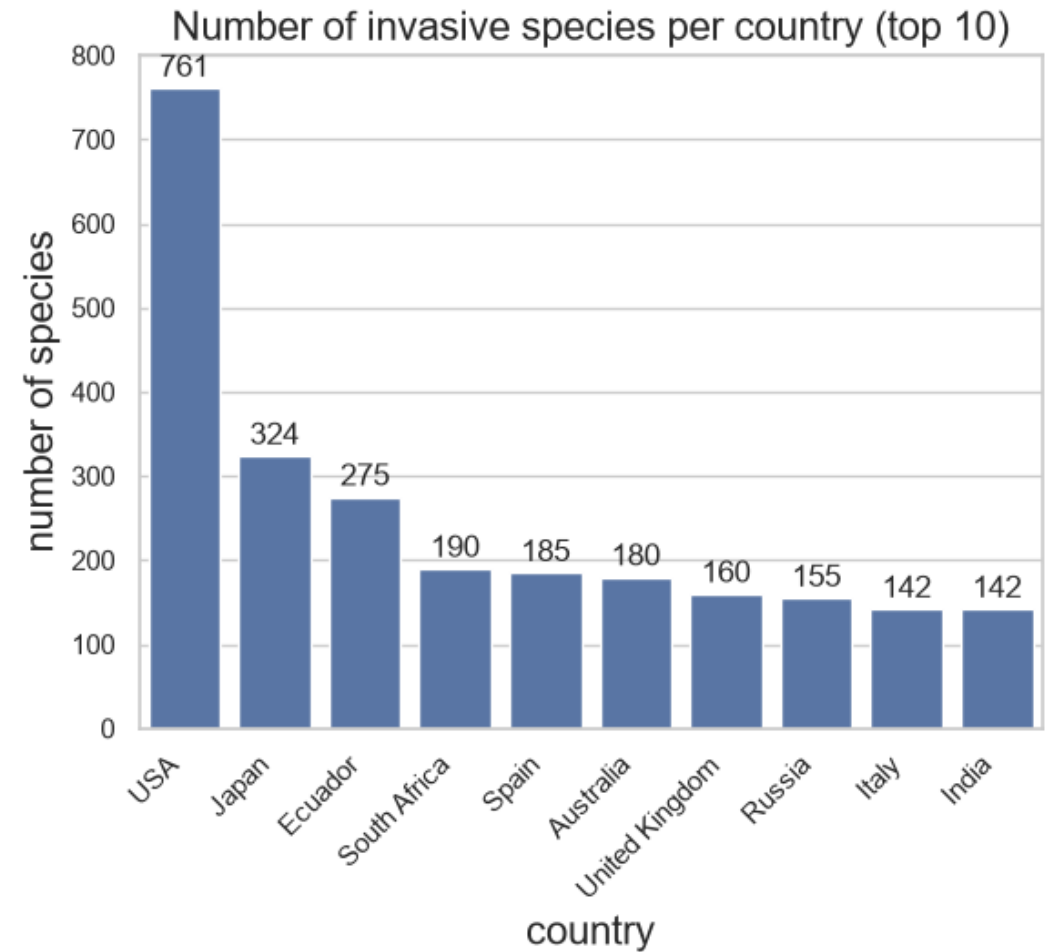
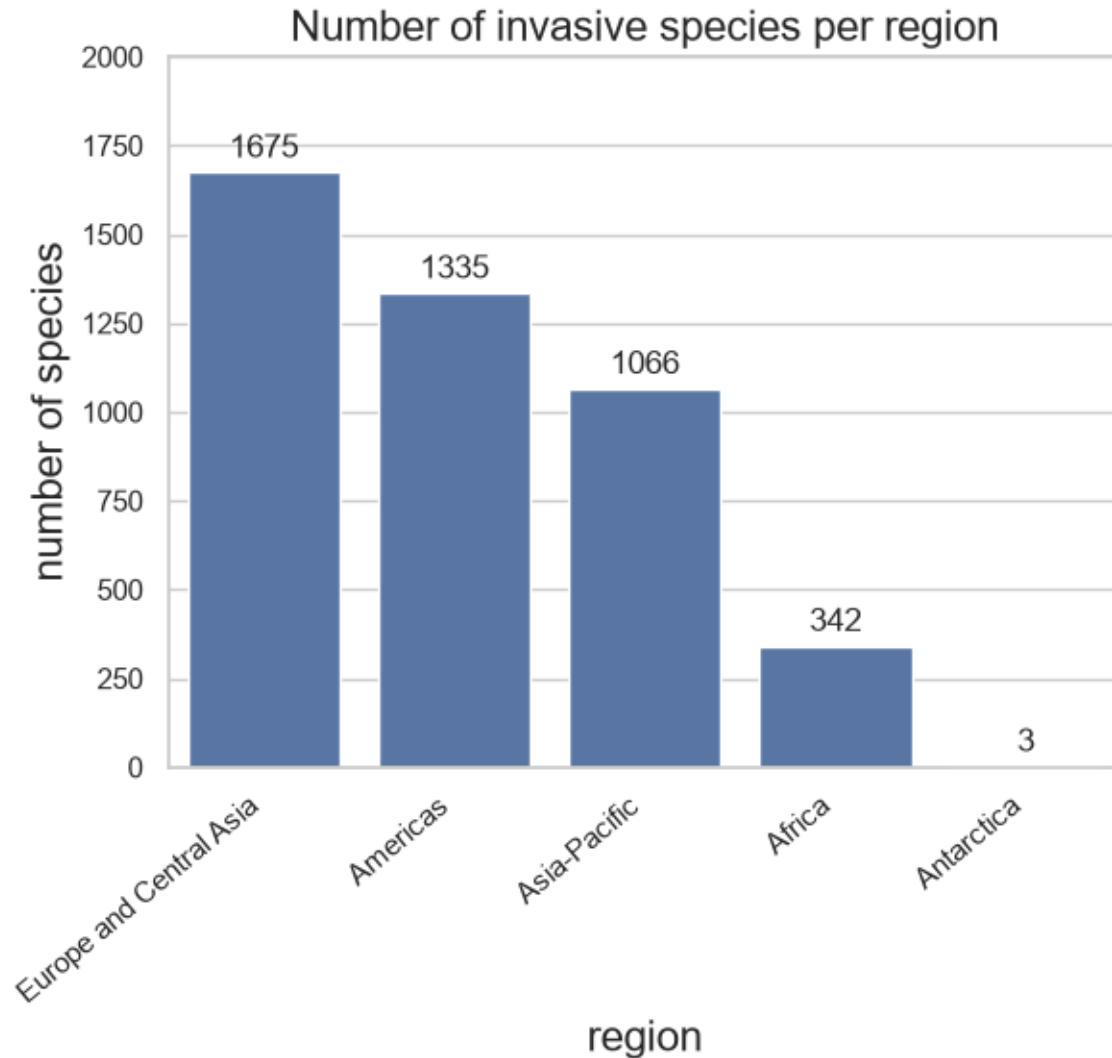
1. When did invasive species introductions peak and was there a period of rapid increase?
2. Which region and countries are the most affected by invasive species?
3. Which invasive species have the greater impact?

1. When did invasive species introductions peak and was there a period of rapid increase?

- Increase since 1980
- Rapid increase since 1990
- Highest number of species recorded: 2010-2019

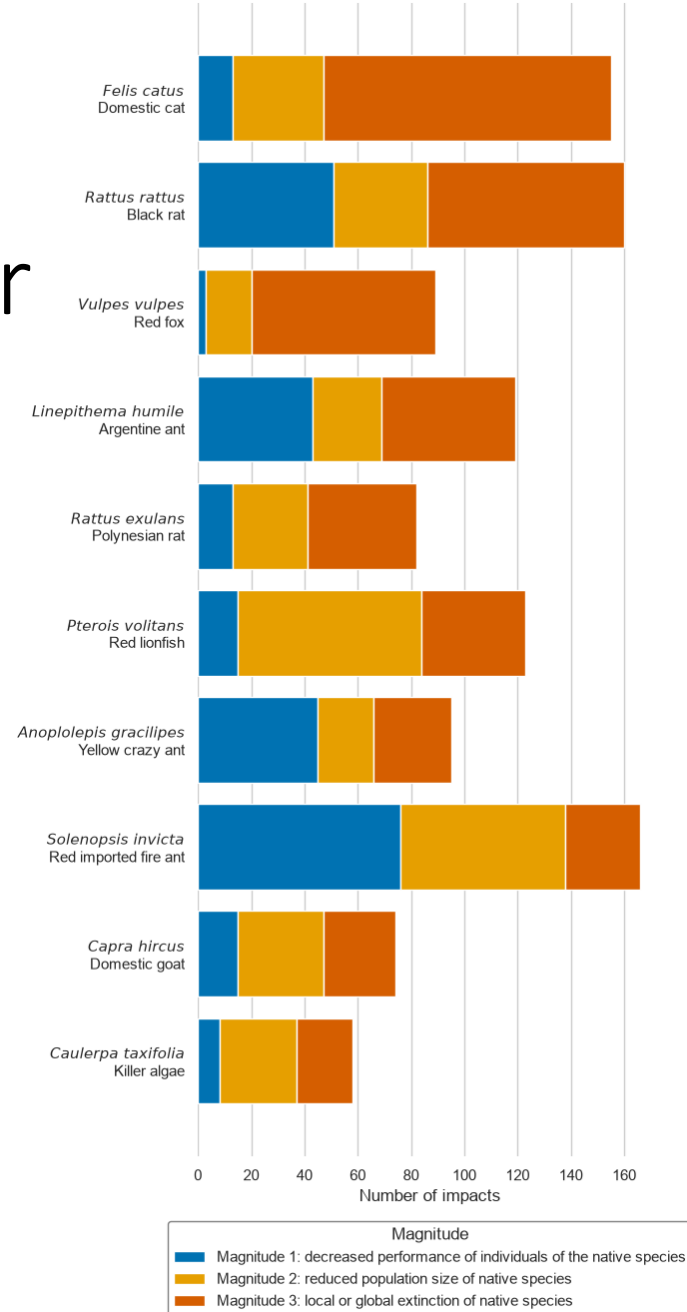


2. Which region and countries are the most affected by invasive species?

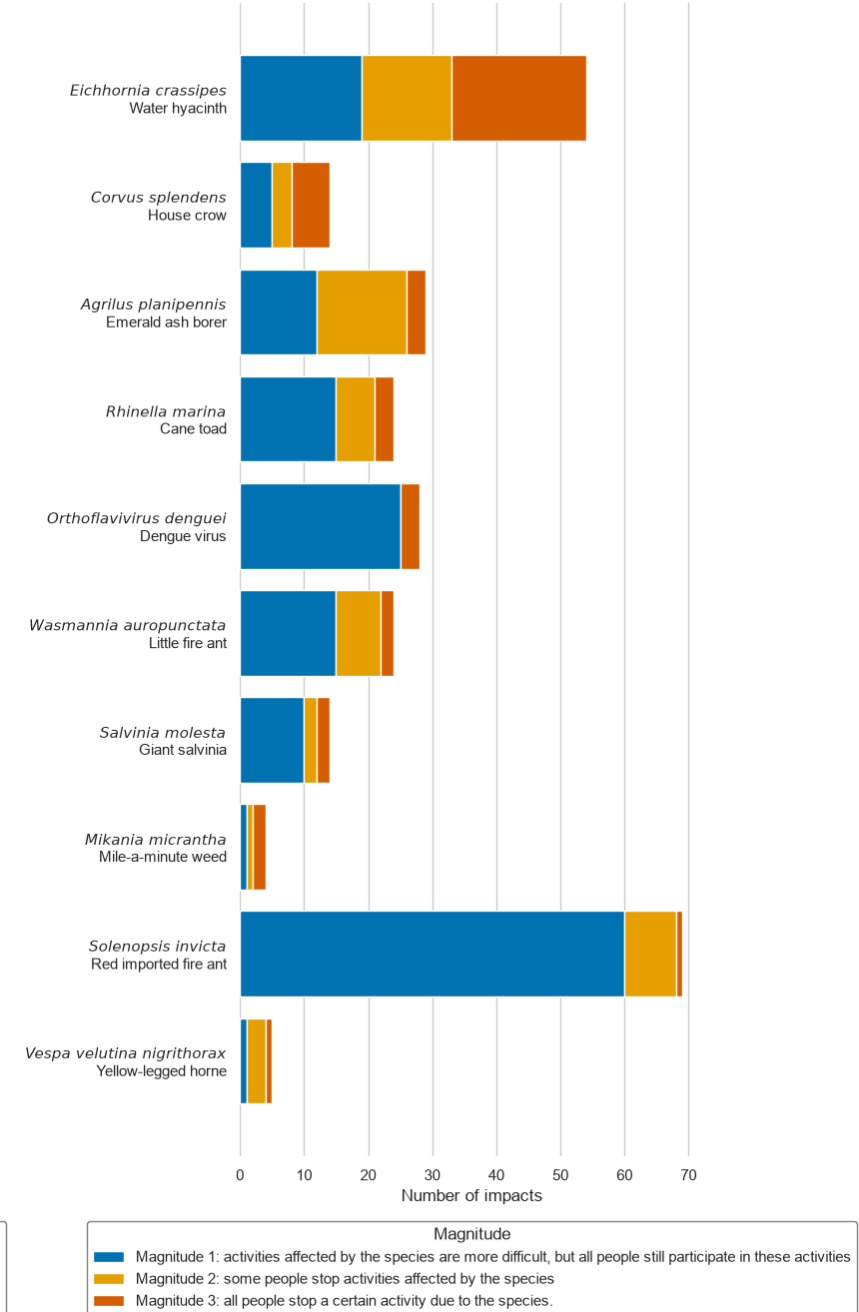


3. Which invasive species have the greater impact?

Species with the greatest impact on the environment

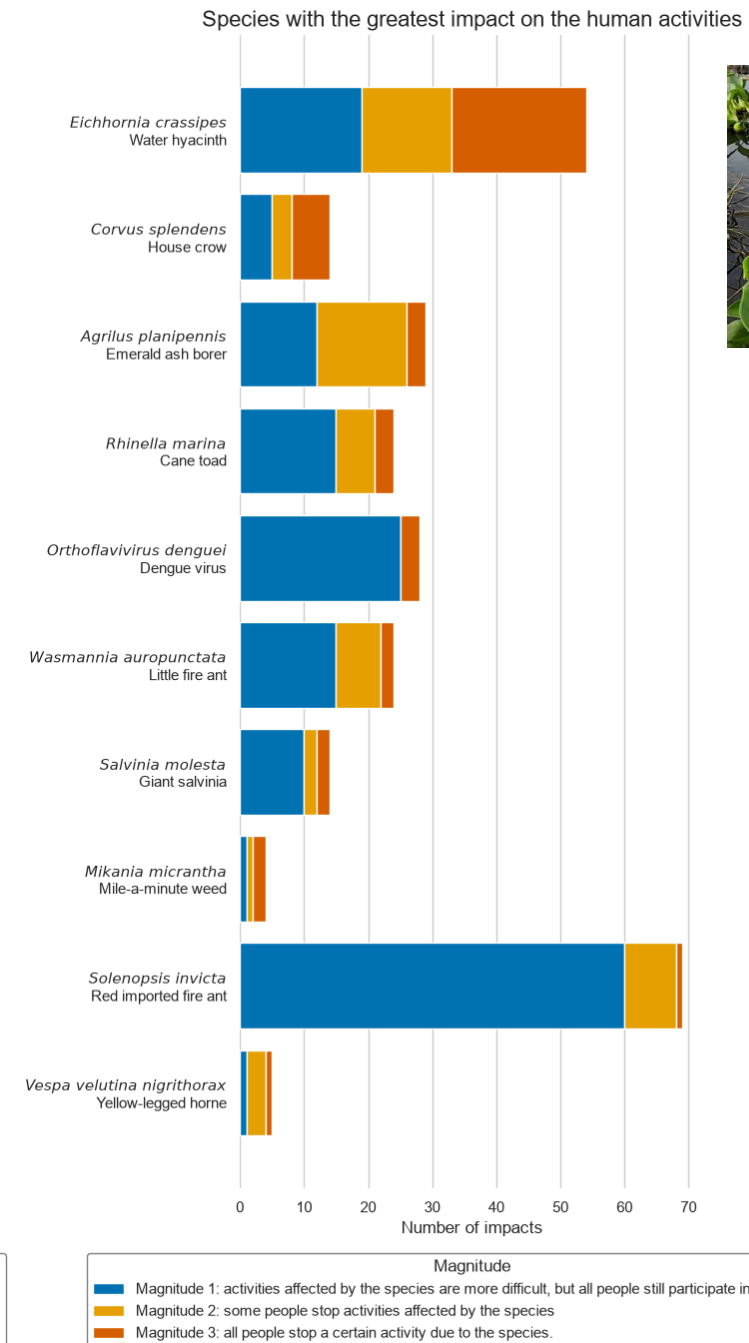
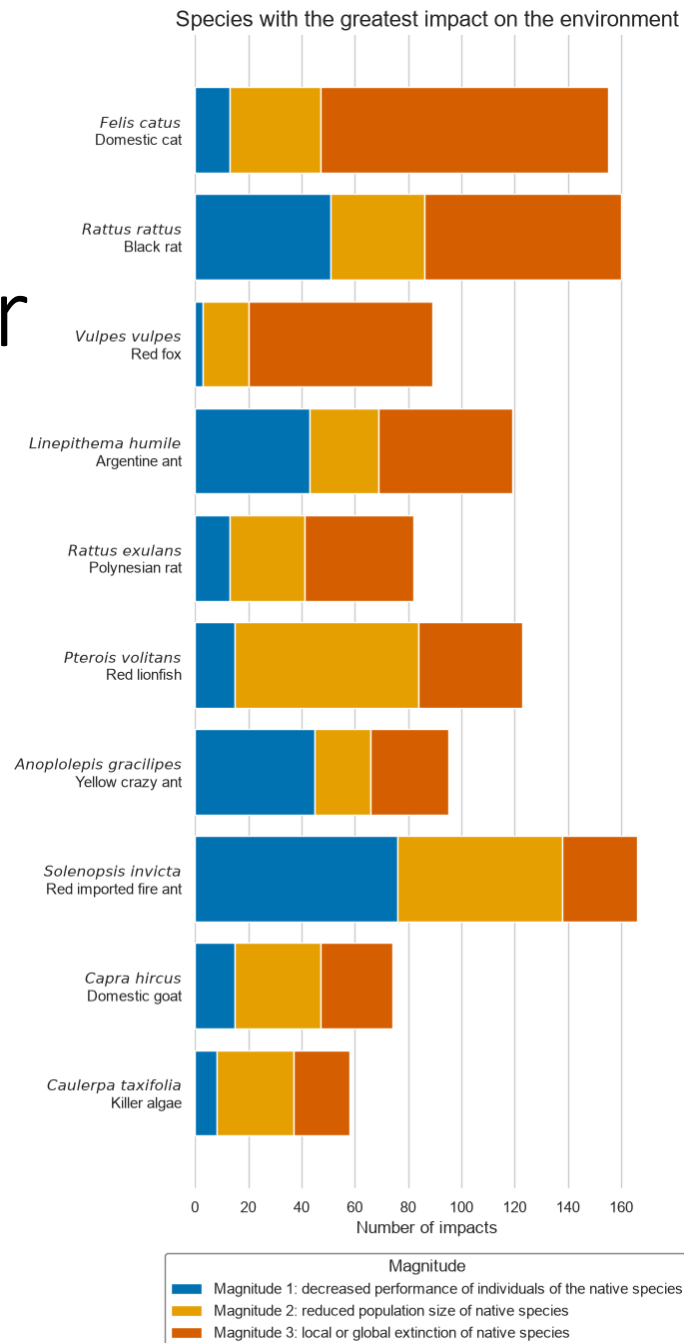


Species with the greatest impact on the human activities



3. Which invasive species have the greater impact?

- Environment
 - greater impact
 - more recorded impacts
 - higher magnitude
- Human activities
 - more magnitude 1 impacts



Conclusions

- Clear increase in number of invasive species recorded
- Environment is more impacted than human activities
- Still unclear:
 - The exact number of invasive species per country

What to do next:

- Achieve a cleaner data set
 - In which countries are the most invasive species?
- Creating a map with a colour gradient depending on the impact of a species on a country

Do you have any questions?

Feel free to ask me later!



Project completed during the

Data Science and Machine Learning Bootcamp
at **Ironhack**

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Technical assistant: Diana